

1 Cross Norms

Notation 1.1. In this section, A_1, A_2 will denote Banach spaces. The norms on these spaces will be denoted by $\|\cdot\|$ unambiguously as we will distinguish between elements of A_1 and A_2 .

Notation 1.2. For Banach spaces A_1, A_2 , we will denote their algebraic tensor product by $A_1 \otimes A_2$. Its completion in some norm $\|\cdot\|_\beta$ will be denoted by $A_1 \otimes_\beta A_2$.

Definition 1.3. A norm $\|\cdot\|_c$ on $A_1 \otimes A_2$ is called a cross norm if it satisfies $\|x_1 \otimes x_2\|_c = \|x_1\| \|x_2\|$ for all $x_1 \in A_1, x_2 \in A_2$.

Remark 1.4. That $\|\cdot\|_\lambda$ is a cross norm is immediate from an application of the Hahn Banach Theorem.

Suppose

Definition 1.5. The injective cross norm $\|\cdot\|_\lambda$ on $A_1 \otimes A_2$ is defined by

$$\|\xi\| = \sup \left\{ \left| \sum_{k=1}^n f(x_{1,k}) g(x_{2,k}) \right| : \begin{array}{l} \xi = \sum_{k=1}^n x_{1,k} \otimes x_{2,k} = \xi, \\ f \in A_1^*, \|f\| = 1, \\ g \in A_2^*, \|g\| = 1 \end{array} \right\}.$$

Remark 1.6. Let $\|\cdot\|_\beta$ be a cross norm on $A_1 \otimes A_2$. Retaining the notation of the previous definition,

$$|f(x_1)| |g(x_2)| \leq \|x_1\| \|x_2\| = \|x_1 \otimes x_2\|_\beta.$$

So, the injective cross norm is the smallest cross norm on $A_1 \otimes A_2$.

Definition 1.7. The projective cross norm on $A_1 \otimes A_2$ is denoted by $\|\cdot\|_\gamma$ and defined as

$$\|\xi\| = \inf \left\{ \sum_{k=1}^n \|x_{1,k}\| \|x_{2,k}\| : \xi = \sum_{k=1}^n x_{1,k} \otimes x_{2,k} \right\}.$$

Remark 1.8. Let $\|\cdot\|_\beta$ be a cross norm on $A_1 \otimes A_2$. For ξ in $A_1 \otimes A_2$, and $\epsilon > 0$, there exists a representation $\xi = \sum_{k=1}^n x_{1,k} \otimes x_{2,k}$ in $A_1 \otimes A_2$, such that

$$\|\xi\|_\beta \leq \sum_{k=1}^n \|x_{1,k} \otimes x_{2,k}\|_\beta = \sum_{k=1}^n \|x_{1,k}\| \|x_{2,k}\| \leq \|\xi\|_\gamma + \epsilon.$$

So, the projective norm majorises all cross norms. That it is itself a cross norm is evident from the computation

$$\|x_1\| \|x_2\| = \|x_1 \otimes x_2\|_\gamma \leq \|x_1 \otimes x_2\|_\lambda \leq \|x_1\| \|x_2\|.$$

projective norm

Observations

1. The injective cross norm is indeed a cross norm on $A_1 \otimes A_2$.

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